

Lecture (3): Quantum Numbers

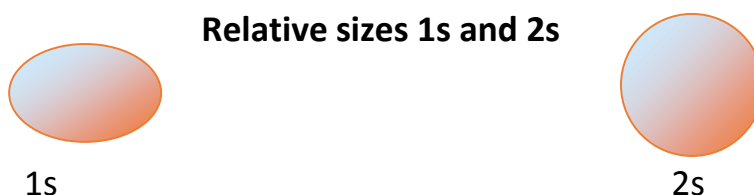
- Quantum numbers are used to differentiate between electrons.
- In quantum theory, each electron in an atom is assigned a set of **four quantum numbers**.
- Three of these give the location of the electron, and the fourth gives the orientation of the electron within the orbital.

Quantum Number: Four quantum numbers are

- 1- Principal Quantum Number (n)
- 2- Angular Momentum Quantum (Azimuthal) (l)
- 3- Magnetic Quantum Number (m_l)
- 4- Spin Quantum Number (m_s)

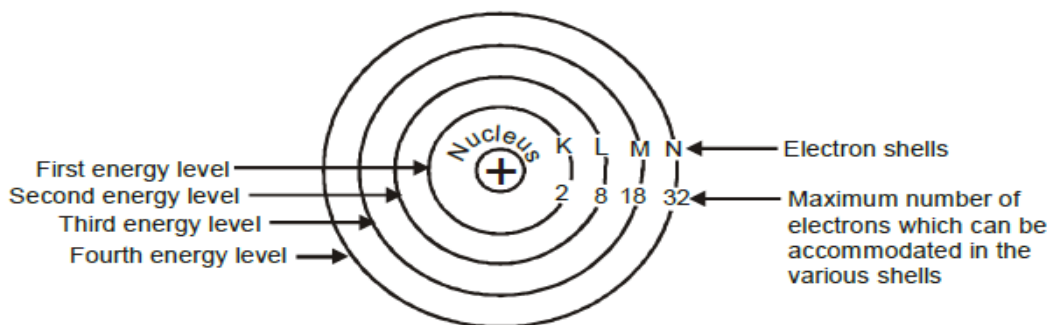
1- Principle Quantum Number (n)

This number describes the energy level that the electron occupies. It can have a value of 1-7 – This defines the “level” of the electron.

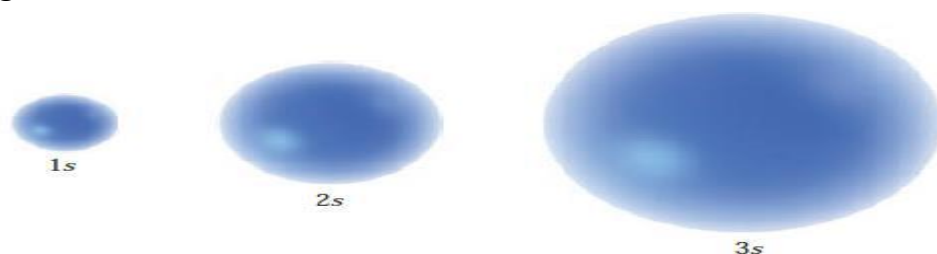


1. The principal quantum number, n , describes the **energy level** on which the orbital resides. and tells the average relative distance of the electron from the nucleus;
2. Every shell is given a specific alphabetic name. The values of n are integers > 0 . First shell ($n = 1$) is known as K shell, second shell ($n = 2$) is known as L shell, third shell ($n = 3$) is known as M shell and so on.
3. Electrons with higher values of n are easier to remove from an atom.
4. Maximum number of electrons in a shell is given by $2n^2$.

Shell	n	$2n^2$	max. no. of electrons
K	1	$2(1)^2$	2
L	2	$2(2)^2$	8
M	3	$2(3)^2$	18
N	4	$2(4)^2$	32



- 5- Each energy level is further divided into subshells designated as s,p,d,f .
 - 1st shell (K) contains 1 subshell (s)
 - 2nd shell (L) contains 2 subshells (s,p)
 - 3rd shell (M) contains 3 subshells (s,p,d)
 - 4th shell (N) contains 4 subshells (s,p,d,f).
- 6- Shells are divided into sub-shells, sub shells further contain orbitals.
 - An orbital may be defined as “A region in the three - dimensional space around the nucleus where the probability of finding the electron is maximum.”
 - The maximum capacity of each orbital is two electrons.
5. Largest E difference is between E levels



The size of the orbital increases as the principal energy level increase

Note : The maximum number of orbitals that can be present in a shell is given by the formula n^2 .

Types of orbitals :

- (1) s-orbitals : The s-subshell contains just one orbital which is non-directional & spherically symmetrical in shape. The maximum number of electrons which can be accommodated in s-orbital is 2.
- (2) p - orbitals : The p-subshell contains three orbitals which have dumb-bell shape and a directional character. The three p-orbitals are designated as p_x , p_y & p_z which are oriented in the perpendicular axis (x,y,z). The maximum number of electrons which can be accommodated in the p subshell is 6 (2 electrons in each of three orbitals).
- (3) d - orbitals : The d-subshells contains 5 orbitals which are double dumb-bell in shape. These orbitals are designated as dxz , dxy , d_{yz} , $d_{x^2-y^2}$, d_{z^2} . The d-subshell can accommodate a maximum of 10 electrons.
- (4) f-orbitals : The f-subshell contains 7 orbitals which are complex in structure. The f-subshell can accommodate a maximum of 14 electrons.

Note : Letters s, p, d & f have originated from the words *sharp*, *principal*, *diffused* & *fundamental* respectively

2- Angular Momentum (Azimuthal) Quantum Number (l)

- This number describes the shape of the orbital that the electron is found in.

- It is represented by the letter ' ℓ '. Sublevel $\ell = 0, 1, 2, 3, \dots n-1$
- It can have a value from 0-3. This defines the "sublevel" of the electron.

n	ℓ
1	0
2	0, 1
3	0, 1, 2
4	0, 1, 2, 3
5	0, 1, 2, 3, 4

ℓ	0	1	2	3	4	5
Name of orbital	<i>s</i>	<i>p</i>	<i>d</i>	<i>f</i>	<i>g</i>	<i>h</i>

- Also, the numbers can be replaced by letters according to the following:

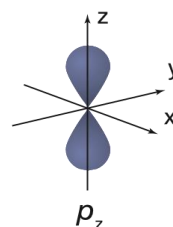
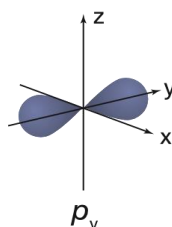
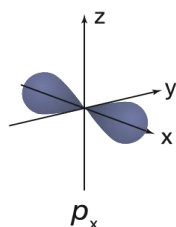
0 = s, 1 = p, 2 = d, 3 = f, 4 = g, h = 5



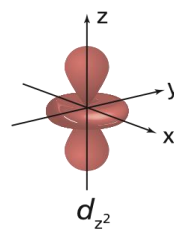
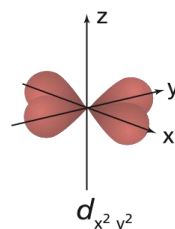
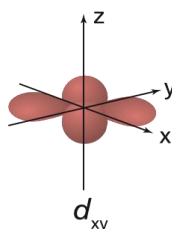
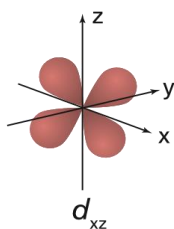
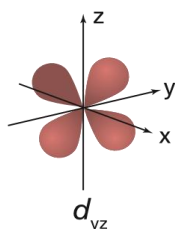
Shape of the orbital s orbital



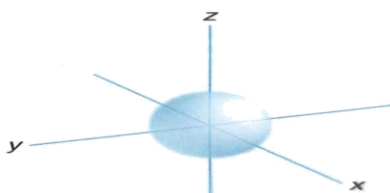
p Orbital Shapes



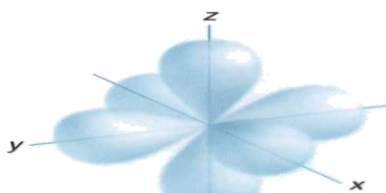
d Orbital Shapes



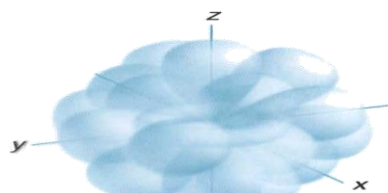
© nfobase Pub sh ng



A
s orbitals:
hold 2 electrons
(outer orbitals of
Groups 1A and 2A)



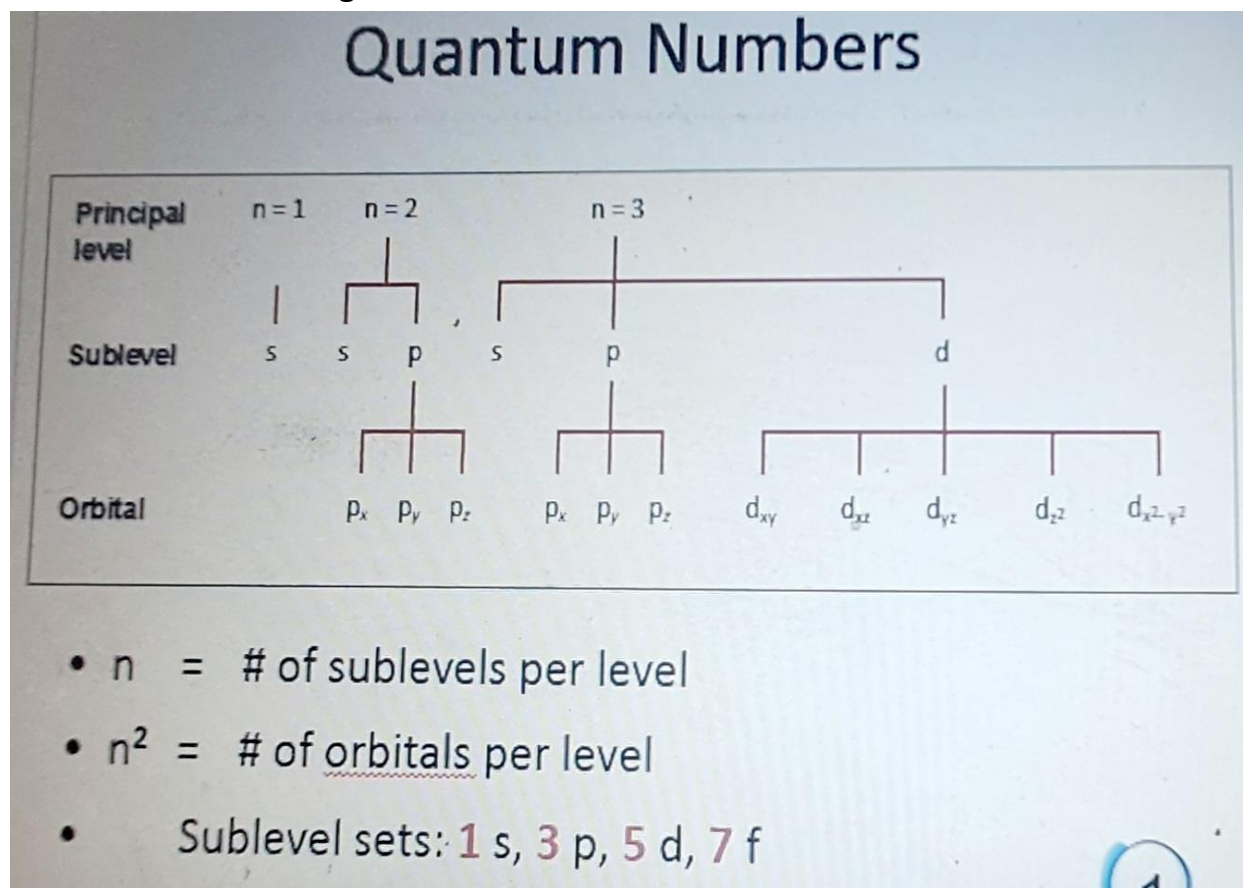
B
p orbitals:
each of 3 pairs of
lobes holds 2 electrons
= 6 electrons
(outer orbitals of
Groups 3A to 8A)



C
d orbitals:
each of 5 sets of
lobes holds 2 electrons
= 10 electrons
(found in elements
with atomic no. of 21
and higher)

- Regions of space occupied by electrons in the same subshell have the same shape but are oriented differently in space.

Quantum Numbers diagram



Maximum Capacities of Subshells and Principal Shells

n	1	2	3	4	...	n
l	0	0, 1	0, 1, 2	0, 1, 2, 3		
Subshell designation	s	s, p	s, p, d	s, p, d, f		
Orbitals in subshell	1	1, 3	1, 3, 5	1, 3, 5, 7		
Subshell capacity	2	2, 6	2, 6, 10	2, 6, 10, 14		
Principal shell capacity	2	8	18	32		$2n^2$

Note : Azimuthal quantum number is also called angular quantum number, - subsidiary quantum number or secondary quantum number

3- Magnetic Quantum Number (m_ℓ):

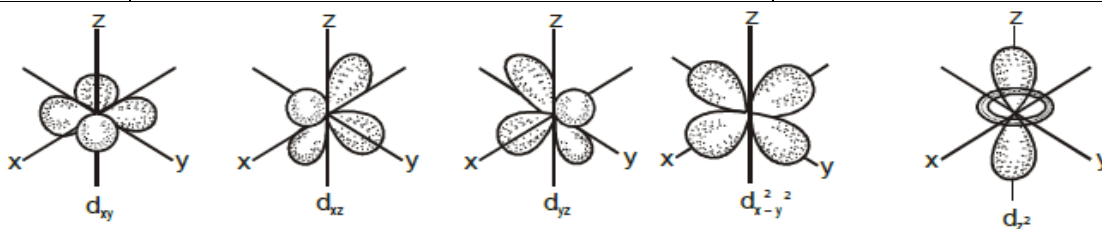
This number describes the orientation of the electrons in the orbitals. This defines the “orbital” of the electron.

a- This quantum number can have the following values: ($-l$ to $+l$) m_l can range from $-l$ to l in integral steps

$$m_\ell = -\ell, (-\ell + 1), \dots, 0, \dots, (+\ell - 1), +\ell$$

- If $l = 0$, m_l can equal 0
- If $l = 1$, m_l can equal $-1, 0, +1$
- If $l = 2$, m_l can equal $-2, -1, 0, +1, +2$
- If $l = 3$, m_l can equal $-3, -2, -1, 0, +1, +2, +3$
- There are $2l+1$ orbitals in each sublevel. Maximum number of orbitals in a sub-shell = $(2\ell + 1)$

Subshell	Orbitals	m
s ($\ell = 0$)	$(2 \times 0 + 1) = 1$	0
p ($\ell = 1$)	$(2 \times 1 + 1) = 3$	-1, 0, +1
d ($\ell = 2$)	$(2 \times 2 + 1) = 5$	-2, -1, 0, +1, +2
f ($\ell = 3$)	$(2 \times 3 + 1) = 7$	-3, -2, -1, 0, +1, +2, +3

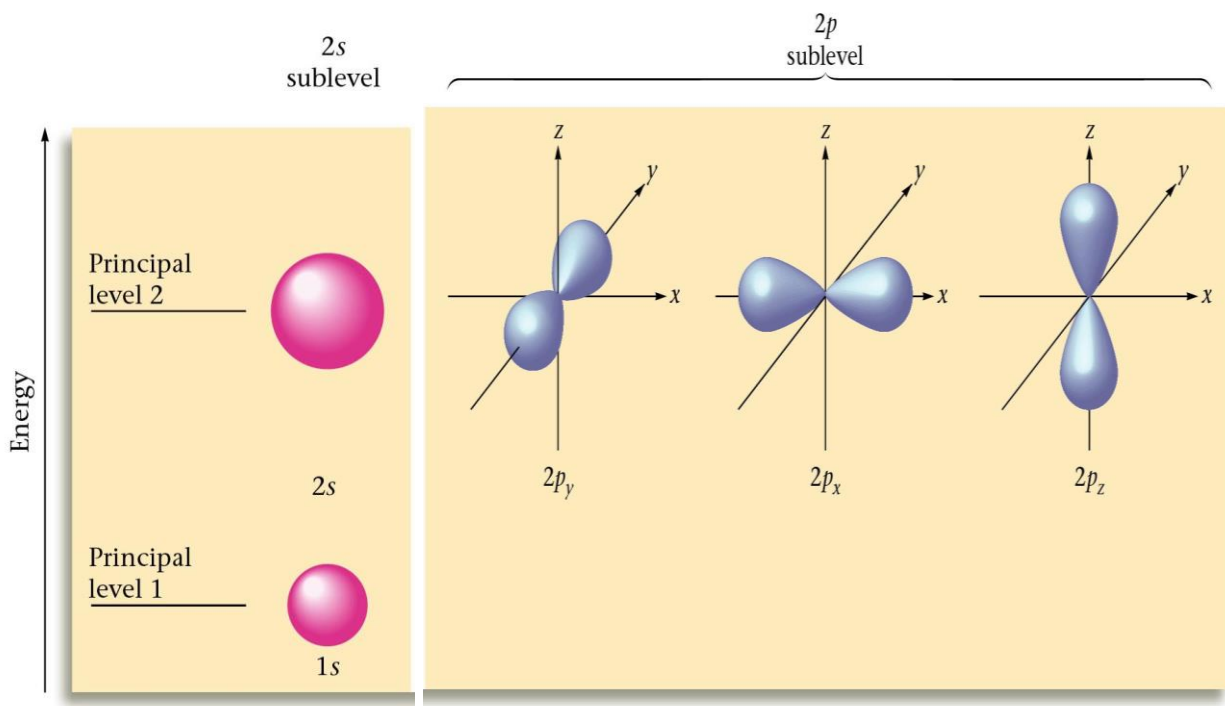


There are seven f-orbitals designated as $f_{yz^2}, f_{xz^2}, f_{z^3}, f_{x(x^2-y^2)}, f_{y(x^2-y^2)}, f_{z(x^2-y^2)}$, and f_{xyz} . Their shapes are complicated ones.

b- Each wave function with an allowed combination of n , l , and m_l values describes an **atomic orbital**, a particular spatial distribution for an electron.

c- Magnetic quantum number (m_ℓ) is describes the orientation of the orbital in space. $-\ell, (-\ell + 1), \dots, 0, \dots, (+\ell - 1), +\ell$

ℓ	m_ℓ	Atomic orbital designations
0	0	s
1	+1, 0, -1	p_x, p_y, p_z
2	+2, +1, 0, -1, -2	$d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$



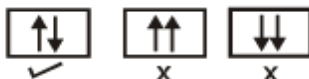
Principal Energy Levels 1 and 2

4- Spin Quantum Number (m_s)

This number describes the direction of spin of the electron in the orbital – electrons in the same level and sublevel must spin in opposite directions. This can have a value of $+1/2$ or $-1/2$ only.

- Electron spin $\Rightarrow +1/2$ or $-1/2$

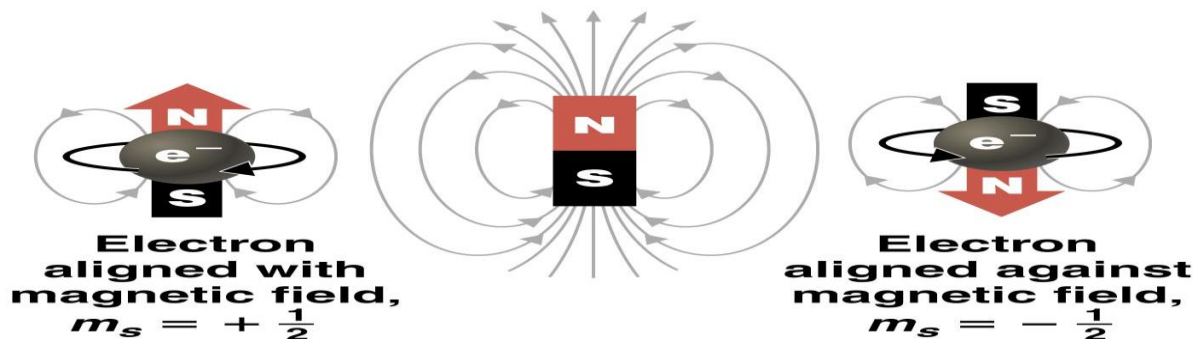
An orbital can hold 2 electrons that spin in opposite directions. **clockwise or counterclockwise. Electron acts like a tiny magnet.** Clockwise spin is represented by $+1/2$ or $\uparrow$ and anticlockwise by $-1/2$ or $\downarrow$



Electron Spin: The Fourth Quantum Number

- When an electrically charged object spins, it produces a magnetic moment parallel to the axis of rotation and behaves like a magnet.
- A magnetic moment is called **electron spin**.

- An electron has two possible orientations in an external magnetic field, which are described by a fourth quantum number m_s .
- For any electron, m_s can have only two possible values, designated + (up) and – (down), indicating that the two orientations are opposite and the subscript s is for spin.
- An electron behaves like a magnet that has one of two possible orientations, aligned either with the magnetic field or against it.



allowable Quantum numbers for the first four Energy shells				
principal Quantum number (n)	angular momentum Quantum number (ℓ)	orbital shape designation	magnetic Quantum number (m_ℓ)	number of orbitals
1	0	1s	0	1
2	0	2s	0	1
	1	2p	-1, 0, +1	3
3	0	3s	0	1
	1	3p	-1, 0, +1	3
	2	3d	-2, -1, 0, +1, +2	5
4	0	4s	0	1
	1	4p	-1, 0, +1	3
	2	4d	-2, -1, 0, +1, +2	5
	3	4f	-3, -2, -1, 0, +1, +2, +3	7
limits of Quantum numbers				
$n = 1, 2, 3 \dots$	$\ell = 0, 1, \dots (n - 1)$	$m_\ell = -\ell, \dots, 0, \dots, +\ell$		

Electronic configuration :

- The maximum number of electrons which can be present in any shell of an atom is given by the formula $2n^2$, where n is the number of shells as counted from the nucleus. Thus, according to this scheme the distribution of the electrons in the different shells is shown in the table given below:

Shell number of energy level	Maximum number of electrons
First shell or K-shell	$2(n)^2 = 2 (1)^2 = 2$
Second shell or L-shell	$2(n)^2 = 2 (2)^2 = 8$
Third shell or M-shell	$2(n)^2 = 2 (3)^2 = 18$
Fourth shell or N-shell	$2(n)^2 = 2 (4)^2 = 32$

- The outermost shell (also called valence shell) cannot have more than 8 electrons and the shell next to it cannot have more than 18 electrons. However, the outermost shell of an atom is the first shell or K shell, then it can not accommodate more than 2 electrons.
- It is not necessary for a given shell to complete itself, before another shell starts forming.
- As a rule, new shell is formed as soon as the shell attains 8 electrons.
- An atom becomes stable (i.e. it stops reacting with other elements), when its outermost shell has 8 electrons or it has only one shell containing 2 electrons.

RULES FOR FILLING OF ORBITALS IN AN ATOM : The filling of orbitals in the ground state is determined by the following values :

1- AUFBAU PRINCIPLE:

- It is a German word, meaning 'building up'
- According to this principle, "In the ground state, the atomic orbitals are filled in order of increasing energies", i.e. in the ground state the electrons occupy the lowest orbitals available to them.
- In fact the energy of an orbital is determined by the quantum number n and ℓ with the help of $(n + \ell)$ rule or Bohr Bury rule.
- This also could be written: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p.

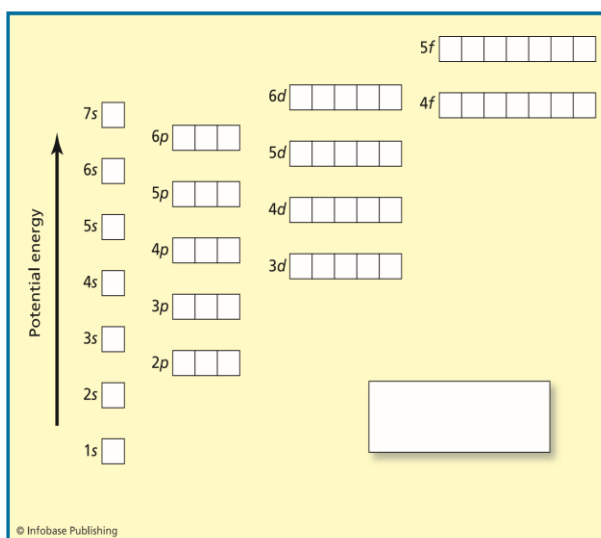
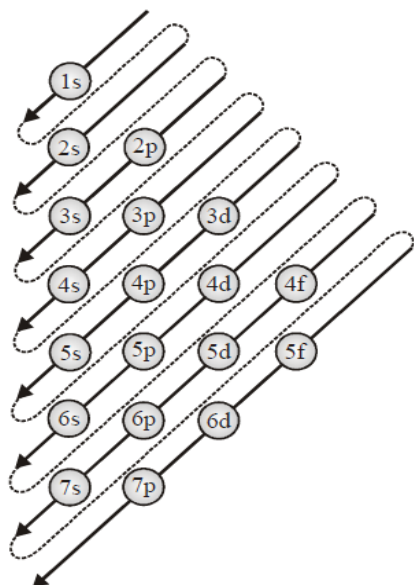


Figure 4.3 Energy levels of the atomic orbitals.

2- Pauli Exclusion Principle

According to the **Pauli Exclusion Principle** :No two electrons in an atom can have the same 4 quantum numbers n, l, m and s identical

(a) An orbital cannot have more than two electrons.

(b) If an orbital has two electrons, they must have opposite spin i.e., they must be paired electrons.

The quantum of Carbon atom C_6

N	No. of electron	L	(m_l)	m_s
1	1	0	0	+1/2
1	2	0	0	-1/2
2	3	0	0	+1/2
2	4	0	0	-1/2
2	5	1	+1	+1/2
2	6	1	0	+1/2

3- HUND'S RULE OF MAXIMUM MULTIPLICITY

(a) This rule governs the filling up of degenerate orbitals of the same sub-shell.

(b) According to this rule "As far as possible electron in the degenerate orbitals remain single, pairing will not take place unless and until each degenerate (equal energy) orbital has got a single electron. For orbital diagrams, you must fill in orbitals in the same energy level with one electron each before pairing up any electrons.

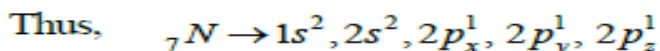
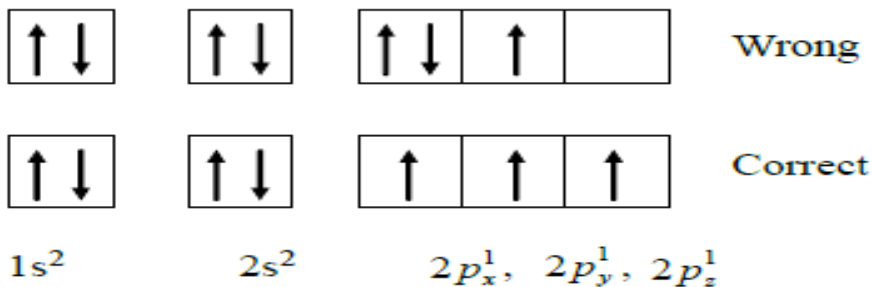
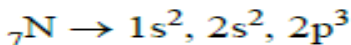
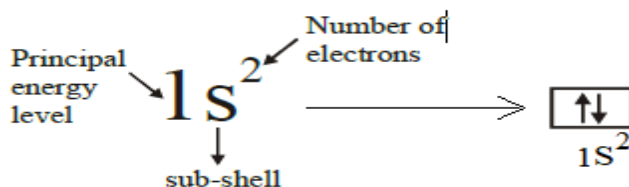


TABLE 4.5 ORBITAL NOTATIONS

CHEMICAL SYMBOL	ATOMIC NUMBER	ORBITAL NOTATION			ELECTRON CONFIGURATION NOTATION
		1s	2s	2p	
H	1	1			1s ¹
He	2	1↓			1s ²
Li	3	1↓	1		1s ² 2s ¹
Be	4	1↓	1↓		1s ² 2s ²
B	5	1↓	1↓	1	1s ² 2s ² 2p ¹
C	6	1↓	1↓	1 1	1s ² 2s ² 2p ²
N	7	1↓	1↓	1 1 1	1s ² 2s ² 2p ³
O	8	1↓	1↓	1↓ 1 1	1s ² 2s ² 2p ⁴
F	9	1↓	1↓	1↓ 1↓ 1	1s ² 2s ² 2p ⁵
Ne	10	1↓	1↓	1↓ 1↓ 1↓	1s ² 2s ² 2p ⁶
MAXIMUM ELECTRONS IN ORBITALS AT A PARTICULAR SUBLEVEL					
s = 2 (one orbital)					
p = 6 (three orbitals)					
d = 10 (five orbitals)					
f = 14 (seven orbitals)					

Atomic Number	Symbol of Element	Electron Configuration	Atomic Number	Symbol of Element	Electron Configuration
1	H	1s ¹	16	S	[Ne]3s ² 3p ⁴
2	He	1s ²	17	Cl	[Ne]3s ² 3p ⁵
3	Li	[He]2s ¹	18	Ar	[Ne]3s ² 3p ⁶
4	B	[He]2s ²	19	K	[Ar]4s ¹
5	Be	[He]2s ² 2p ¹	20	Ca	[Ar]4s ²
6	C	[He]2s ² 2p ²	21	Sc	[Ar]4s ² 3d ¹
7	N	[He]2s ² 2p ³	22	Ti	[Ar]4s ² 3d ²
8	O	[He]2s ² 2p ⁴	23	V	[Ar]4s ² 3d ³
9	F	[He]2s ² 2p ⁵	24	Cr	[Ar]4s ¹ 3d ⁵
10	Ne	[He]2s ² 2p ⁶	25	Mn	[Ar]4s ² 3d ⁵
11	Na	[Ne]3s ¹	26	Fe	[Ar]4s ² 3d ⁶
12	Mg	[Ne]3s ²	27	Co	[Ar]4s ² 3d ⁷
13	Al	[Ne]3s ² 3p ¹	28	Ni	[Ar]4s ² 3d ⁸
14	Si	[Ne]3s ² 3p ²	29	Cu	[Ar]4s ¹ 3d ¹⁰
15	P	[Ne]3s ² 3p ³	30	Zn	[Ar]4s ² 3d ¹⁰